Rohan Bali

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Research Interests Reliable machine learning for Earth observation: knowing when a model's measured skill, its explanation of failure, and its uncertainty can be trusted in places and periods it was not trained on. I find evaluation confounds that masquerade as architectural differences, test whether cross-region explanations are identifiable at all when covariate overlap is thin, and measure how well deep-learning uncertainty tracks real error under distribution shift.

Education

University of Massachusetts Dartmouth North Dartmouth, MA, USA
M.S., Data Science, Department of Computer & Information Science Sep 2024 – May 2026
GPA: 3.77/4.0 (3.86 in CS, data science and mathematics courses)
Thesis: *Multi-Horizon Urban Growth Prediction: A Comparative Deep Learning Study* (Advisor: Dr. Donghui Yan)
Selected coursework: Advanced Mathematical Statistics (MTH 522), A; Small World Networks (MTH 565), A; Advanced Machine Learning (CIS 550), A−; High Performance Scientific Computing (DSC 520), A−

Manipal University Jaipur Jaipur, India
B.Tech., Computer and Communication Engineering Jul 2017 – Sep 2021
CGPA: 6.30/10 · Final-year GPA: 8.53/10 (semesters 7–8), with an A in Machine Learning
Minor project: *Road Safety Analysis using Machine Learning* (Supervisor: Dr. Vijaypal Singh Dhaka)

Publications *All sole-authored, with public code and data.*

[1] **Bali, R.** (2026). Channel count, not horizon length, drives architectural divergence in multi-horizon urban growth prediction. In *Proceedings of the 1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026)*, Ghent, Belgium. **Published; oral presentation.**
Shows that losing input channels, not lengthening the forecast horizon, drives most of the apparent architectural divergence in urban-growth prediction.
Paper (Zenodo) · Code and benchmark

[2] **Bali, R.** (2026). The channel-count confound: A continental audit of multi-horizon urban growth prediction. In *Spatial Data and Intelligence: 7th International Conference, SpatialDI 2026*, Changsha, China. Lecture Notes in Computer Science, Springer. **Accepted, to appear.**
Extended version of [1], with the full ablations and the MC Dropout uncertainty analysis.
Code, benchmark and pre-trained weights

[3] **Bali, R.** (2026). Confident but unidentifiable: Positivity limits on cross-region attribution in geospatial models. *Transactions in GIS*. **Accepted pending revisions.**
Derives a positivity diagnostic and partial-identification bound for cross-region attribution; across 44 cities, a confident, well-fit explanation gets the sign of the transfer gap wrong in every degrading model.
Code and data

Research Experience

Independent Researcher Jun 2026 – Present

- Revising [3] for *Transactions in GIS* and continuing cross-region transfer research; maintaining public code, data and benchmarks for all papers.

Graduate Researcher (M.S. thesis), UMass Dartmouth Sep 2025 – May 2026
Department of Computer & Information Science · Advisor: Dr. Donghui Yan

- Designed a continental audit of multi-horizon urban-growth prediction: 5,698 CONUS tiles at 250 m built on Google Earth Engine, a sealed 2020 temporal holdout, and zero-shot transfer to 20 Lagos tiles (63,840 pixels).
- Isolated a channel-count confound (24 → 21 input channels) that explains 94% of a CNN's

Figure-of-Merit decline at the longer horizon (delta-method 95% CI 62–100%), an effect prior work had attributed to horizon length.
- Ran 1,352 MC Dropout inference runs over 8.69M pixels, showing uncertainty ranks errors reliably ($r = 0.983$) but understates their size by 1.4–1.9×.
- Published as [1] (GeoAI 2026, oral) and [2] (Springer LNCS), with an open benchmark.

Research Project, UMass Dartmouth Sep 2024 – Dec 2024
Supervisor: Prof. Dana Fine · MTH 565, Small World Networks

- Built a co-authorship network of 33,981 researchers and 624,928 links from OpenAlex (via PyAlex); found a clustering coefficient of 0.71, about 650 times that of a density-matched random graph, with a giant component spanning 61% of authors.
- Wrote my first independent research report: *Understanding Academic Collaboration Networks Through Small-World Theory*.

Undergraduate Research Project, Manipal University Jaipur Jan 2020 – May 2020
Supervisor: Dr. Vijaypal Singh Dhaka · CC1634, Minor Project

- Modeled road-accident hotspots from Indian government accident data, comparing logistic regression with tree-based and other classical classifiers; a solo project and my first machine-learning research.

Professional Experience

Data Scientist, The Parivartan Foundation Feb 2023 – Aug 2024
NGO, Shimla, Himachal Pradesh, India

- Collected and analyzed satellite and government data during the 2023 Himachal Pradesh floods to locate about 20 cut-off communities and help direct relief to them.
- Sourced and analyzed public government datasets to identify community-level priorities for outreach and civic-engagement campaigns.

Software Engineer, Capgemini Dec 2021 – Aug 2023
Mumbai, India

- Supported production infrastructure and job-scheduling workflows; built Python, PowerShell and Bash automation for data and systems operations across cloud environments.

Machine Learning Researcher, Upcred.ai Jun 2021 – Dec 2021
Bangalore, India

- Designed Python and Beautiful Soup web-scraping pipelines and MongoDB schemas supplying training data for the team's machine-learning models.

Presentations

Bali, R. Channel count, not horizon length, drives architectural divergence in multi-horizon urban growth prediction. Oral presentation, *GeoAI 2026*, Ghent, Belgium, Jun 2026.

Honors

National Talent Search Examination (NTSE) Scholar, Government of India 2014
National merit scholarship awarded by NCERT through a two-stage national examination.

Skills

Methods: distribution-shift evaluation (sealed temporal holdouts, zero-shot geographic transfer, source-by-target matrices); identification (transportability, positivity, partial-identification bounds); MC Dropout calibration; delta-method inference.

Models: CNN, U-Net, ConvLSTM, LSTM/GRU, Transformers, geospatial foundation models.

Earth-observation data: Landsat, Sentinel, NLCD, PANGAEA; continental-scale tiling.

Programming: Python (PyTorch, TensorFlow, scikit-learn, NumPy, pandas, xarray, Rasterio, GeoPandas, GDAL), SQL, C++, Bash, PowerShell.

Tools: Git, Docker, LaTeX, Google Earth Engine, MPI/OpenMP (Purdue Anvil HPC).

References

Dr. Donghui Yan, Associate Professor of Mathematics, UMass Dartmouth (M.S. thesis advisor)
Prof. Dana Fine, Department of Mathematics, UMass Dartmouth (research project supervisor)
Dr. Vijaypal Singh Dhaka, Professor, Manipal University Jaipur (B.Tech project supervisor)